

Basil C K

*AI & Data Science Graduate | Python Backend | Machine Learning |
Computer Vision*

basilck618@gmail.com | +91 8111974510 | Portfolio: basilbaasi.github.io | LinkedIn | GitHub

Thrissur, Kerala, India

Profile

B.Tech graduate in Artificial Intelligence & Data Science with strong skills in Python, Machine Learning, Computer Vision, and Backend Development. Hands-on experience building end-to-end ML pipelines, REST APIs, and AI-driven applications. Proficient in data preprocessing, model development, and integrating machine learning models into real-world systems. Familiar with SQL, version control (Git), and software development fundamentals including operating systems and networking. Seeking entry-level roles in AI, Backend Development, and Machine Learning Engineering.

EDUCATION

B.Tech in Artificial Intelligence and Data Science

JCT College of Engineering and Technology, Anna University

2022 – 2026

CGPA: 8.2 / 10

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Database Management Systems, Software Engineering, Machine Learning

TECHNICAL SKILLS

Programming & Backend:

Python (automation, backend logic, ML pipelines), Flask (REST API development)

Machine Learning & Computer Vision:

CNN architectures, YOLOv8 object detection, OpenCV image processing, model training & evaluation

Data & ML Foundations:

Pandas, NumPy, supervised learning fundamentals, SQL

Systems & Networking:

Operating Systems fundamentals, HTTP/ REST architecture, client-server systems, TCP/ IP basics

Cloud & Platforms:

Basic exposure to OCI / AWS fundamentals

Tools & Development:

Git (branching, commits, workflows), Jupyter Notebook, VS Code

Exposure to:

Linux (CLI, permissions, processes), Docker (containerization basics), CI/ CD pipelines (conceptual understanding), ML model deployment workflows

INTERNSHIP EXPERIENCE

Data Analytics Virtual Experience – Accenture (Forage)

- Performed data cleaning, exploratory analysis, and visualization on structured datasets.
- Identified user engagement trends and generated insights for a simulated business scenario.
- Presented findings and recommendations in a structured report format.

PROJECT EXPERIENCE

Hybrid CNN-YOLO Framework for Car Damage Severity Classification and Localization

Tech Stack: Python, YOLOv8, TensorFlow/Keras, PyTorch, OpenCV, Flask, HTML/CSS

- Designed and implemented an **end-to-end computer vision system** combining YOLOv8 object detection and a CNN classifier for automated vehicle damage assessment.
- Built a **multi-stage inference pipeline** where YOLO detects damaged regions in vehicle images and a CNN model classifies each region into **Minor, Moderate, or Severe** damage categories.
- Trained the CNN classifier on **224×224 images**, achieving **87.4% training accuracy and 54.4% validation accuracy**.
- Trained a **YOLOv8 detection model** achieving **mAP@0.5 of 0.659 and mAP@0.5:0.95 of 0.647** for vehicle damage localization on a custom dataset.
- Developed a **Flask-based web application** enabling users to upload images and receive automated damage analysis with bounding box visualization and confidence scores.
- Implemented the full ML workflow including **dataset preparation, model training, inference integration, and deployment-ready application structure**.

Vocalix – AI-Powered Voice Assistant

Tech Stack: Python, Speech Recognition, LLM APIs, Automation Libraries

- Developed a **Python-based voice assistant architecture** capable of handling voice commands and generating AI responses.
- Integrated remote **LLM APIs** for conversational processing.
- Implemented system automation features including **application control and file navigation**.
- Designed structured input → processing → response workflows for interactive user interaction.

Vexilon – AI-Powered Chrome Assistant

Tech Stack: Python, Flask, REST APIs, LLM APIs, Chrome Extension, Git

- Designed and developed a Chrome extension integrated with a **Flask backend AI service**.
- Built **RESTful API endpoints** for request handling and response management.
- Integrated external **LLM APIs (Gemma, Ollama)** using OpenAI-compatible formats.
- Structured the project with **modular backend components and version-controlled workflows**.

ADDITIONAL LEARNING & CERTIFICATIONS

- **IBM Data Science** – Selected Coursework (Python, Data Analysis)
- **Oracle Cloud Infrastructure 2023** – AI Foundations Associate
- **Google AI Essentials**

LANGUAGES

English | Malayalam | Tamil